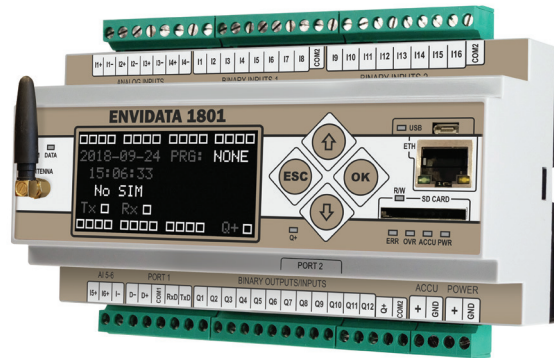


ENVIDATA 1801 Mobile Controller for GSM/GPRS telemetry

- **2G/3G transmission**
- Embedded GSM 2G/3G modem
- Dual-SIM technology – access to 2 independent GSM networks ensures superior availability
- **16 binary inputs** (galvanic isolation)
- **12 binary outputs**, selectively configurable as inputs (galvanic isolation)
- **4 analog inputs 4-20 mA** (galvanic isolation)
- **2 analog inputs 4-20 mA** (w/o galvanic isolation)
- **Ethernet port 10Base-T/100Base-TX**
- RS-232/485 serial port for external devices (galvanic isolation)
- RS-232 port with 5 V feeding for operator panels
- OLED graphic display (128x64 pixels)
- Battery buffered power supply (SLA battery support)
- **Data logger with 0,1 sec resolution (SD card support)**
- Programmable logic controller (PLC)



- Standard communication protocols (MODBUS RTU, MODBUS TCP, M-BUS, NMEA 0183)
- FlexSerial – programmable handling of non-standard serial protocols
- Remote configuration, programming, diagnostics and firmware upgrade via GPRS

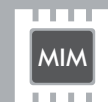
ENVIDATA 1801 is a family of new generation telemetry controller for demanding tasks and applications. ENVIDATA 1801 model is a professional, industrial design combining functionality of programmable logic controller, data logger, protocol converter and wireless communication interface for GPRS packet transmission over GSM network. Dual-SIM technology ensures superior level of GSM network availability, with redundant channel of data transmission. Ethernet port provides powerful capabilities of integration with other devices and systems of the user. Graphic display is a convenient user interface for local diagnostics, supervision and monitoring - without use of external operator panel or portable PC. With compact, robust design, integral GSM modem, attractive technical features and easy to use configuration tools the ENVIDATA 1801 controller is an optimal solution for demanding wireless telemetry, control, diagnostic, surveillance and alarm systems.

Resources

- 16 optoisolated binary/counter inputs 12/24 VDC (I1 - I16), positive and negative logic
- 12 optoisolated binary outputs 12/24 VDC (Q1 - Q12), positive logic – selectively configurable as inputs
- 4 optoisolated, differential analog inputs 4-20 mA (accuracy 0,2%, 14-bit resolution @ 1 sec interval) with configurable hysteresis and filtration
- 2 single-ended analog inputs 4-20 mA
- Ethernet port 10Base-T/100Base-TX
- Isolated RS-232/485 serial port
- RS-232 serial port with 5 V / 500 mA feeding
- micro USB (AB) port for local configuration and programming
- Interface for backup 12 V SLA battery – charging support
- 2 SIM holders - Dual-SIM support
- OLED graphic display (128x64 pixels) and status LEDs
- Internal flags and registers for user application program
- Firmware Flash memory with remote update capability
- Data and Event logger supporting SD card
- RTC with external synchronization functions

Functionality

- Access to module resources using standard protocols MODBUS RTU and MODBUS TCP
- Intelligent packet routing and Multimaster support in MODBUS mode
- Programmable control logic using I/Os, timers, counters, flags and registers for triggering events (data transmission/recording, **SMS transmission**, setting outputs and internal registers)
- Event based transmission (unsolicited messaging) triggered by change of binary input state, internal flag state, by reaching alarm level of analog input, by true condition.
- Configurable SMS messages triggered by alarms and scheduled
- Dynamic fields in SMS text
- Configurable alarm levels, hysteresis, deadband and filtration for analog inputs
- Data and event recording on SD card with 0,1 sec res.
- Transmission of data from external devices connected to RS-232/485 serial port
- 5 V feeding provided for external device connected to RS-232 serial port (e.g. operator panel, GPS receiver)
- Configurable events based on mirrored resources of external devices
- Remote configuration and programming via 2G/3G and Ethernet network
- Configurable access security – list of authorized IPs and tel. numbers, optional password
- DIN rail mounting
- Supply voltage 12/24 VDC (24 VDC required for battery buffered power supply operation)
- Built-in management of external SLA backup battery
- Built-in advanced auto-diagnostics
- Detachable terminal blocks
- Sending logged data as a text or CVS format file to the configured FTP servers



16-28DI
/12DO

6AI



DIN RAIL

RS-232

RS-232/485

3G



General

Dimensions (L x W x H)	157x86x58 mm
Weight	450 g
Fixing	DIN Rail 35 mm
Operating temperature	-20 ... +65 °C
Protection class	IP40

GSM/GPRS Modem

Modem type	Cinterion EHS6
GSM	QuadBand (850/900/1800/1900)
UMTS	five band ranges (800, 850, 900, 1900, 2100)
Antenna	50 Ω

Power supply

DC (nom. 12/24 V)	10,8 – 30 V		
Input current (@ 24 VDC)	Idle 0,06 A	Active 0,25 A	Max. 1,00 A

Inputs I1...I16 *

Input voltage range	0 – 30 V
Input current limit	2,4 mA
Input voltage ON (1)	> 9,4 V
Input voltage OFF (0)	< 8,4 V

Inputs Q1...Q12 *

Maximum input voltage	30 V
Input current limit	2,4 mA
Input voltage ON (1)	> 9,4 V
Input voltage OFF (0)	< 8,4 V

Outputs Q1...Q12

Maximum output current	100 mA
Voltage drop @ 100 mA	< 0,5 V max.
OFF state current	< 100 μA max.

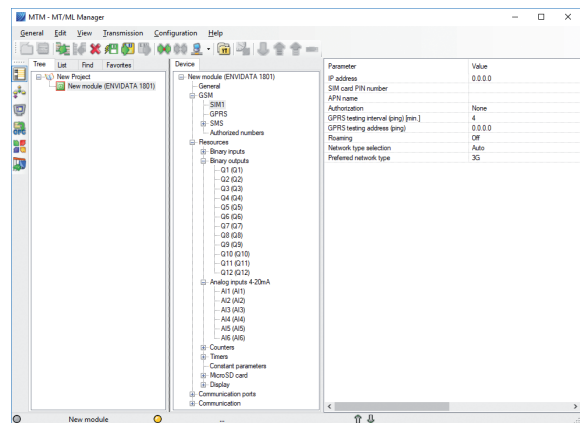
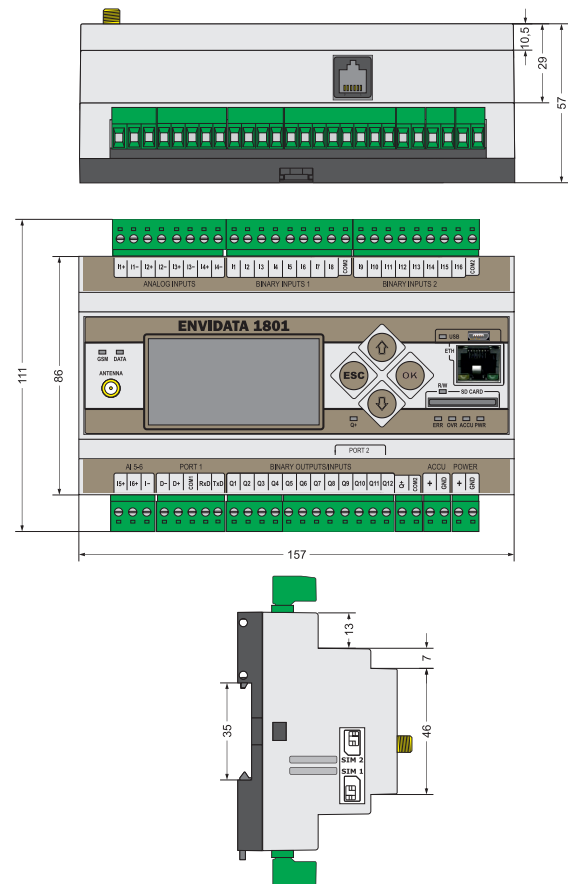
Analog inputs 4-20 mA (4) (I1+ ... I4+)

Input current range	4 – 20 mA
Maximum input current	50 mA
Dynamic input impedance	55 Ω typ.
Voltage drop @ 20 mA	< 5 V
A/D converter resolution	14 bits
Accuracy (@ 25 °C)	0,2 %

Analog inputs 4-20 mA (2) (I5+, I6+)

Input current range	4 – 20 mA
Maximum input current	30 mA
Dynamic input impedance	110 Ω typ.
Voltage drop @ 20 mA	< 2,2 V
Accuracy (@ 25 °C)	0,5 %

* according to IEC 61131-2 for switch type 1 and 3

Configuration utility**Drawings and dimensions (in millimeters)****Additional info:**

inventia

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INVENTIA complies with ISO 9001:2015 certified Quality Management System.

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